Language Models, Embeddings, and Representations

DSA4213 Assignment 1 | Experimental report | 16 October 2026

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1. Questions, data and experimental controls

This report examines whether better next-token prediction also yields more useful sentiment representations. Three comparisons address this question: which model predicts the next token best on a small common corpus; how frozen input embeddings change an LSTM; and whether better language modeling yields better sentiment representations. The strongest observed novel LM is LSTM (PPL 75.66), but its random/trainable variant does not have the highest downstream score. A task-trained TF-IDF baseline exceeds all three frozen LM feature sets.

1.1 Language-model data

The corpus is *Pride and Prejudice*, Project Gutenberg ebook 1342, with front matter, illustration blocks, chapter headings and emphasis underscores removed, retaining 2,154 paragraphs (685,095 characters). A fixed regular expression lowercases text, normalizes curly quotes, and retains words with internal apostrophes, numbers and punctuation. Complete paragraphs are split in original order at approximately 80/10/10% of tokens; no training example crosses a split boundary.

Table 1. Shared novel splits; targets include EOS, while body tokens do not.

Split	Paragraphs	Body tokens	Scored targets	Body UNK
Train	1,674	116,727	118,401	2.01%
Validation	236	14,413	14,649	2.99%
Test	244	14,567	14,811	3.21%

The 3,620-entry vocabulary is built from training counts only: 3,616 ordinary tokens occurring at least twice, plus PAD/UNK/BOS/EOS. Each paragraph supplies input BOS+body and target body+EOS. Loss excludes PAD and never scores BOS; CE is summed negative natural-log probability divided by all valid targets, and PPL=exp(CE). Output distributions retain the full vocabulary. All models use preceding tokens only.

1.2 Models and matched training budget

Table 2. Part I models; neural input embeddings are random and trainable.

Model	Architecture / context	Parameters
n-gram	Interpolated unigram/bigram/trigram; at most two preceding tokens	Count tables
RNN	One tanh layer; embedding 100, hidden 128; full paragraph	858,420
LSTM	One unidirectional layer; embedding 100, hidden 128; full paragraph	946,740
Transformer	One causal pre-LN block; embedding 100 projected to 128; 4 heads; FFN 256; sinusoidal positions	974,644

Each n-gram order is smoothed before interpolation. A search over nine validation configurations selects alpha=0.001 and unigram/bigram/trigram weights 0.3/0.5/0.2. Neural models use Adam, learning rate 0.003, batch 8, gradient clipping 1, seed 42, an eight-epoch cap and validation patience 3. Each run creates a fresh seeded loader and restores the lowest-validation-CE checkpoint. Test data do not select these settings.

Neural models retain all paragraph targets, including the longest 2,824-target training sequence. RNN/LSTM use packed sequences and reset state per paragraph. Transformer uses causal attention, GELU and dropout 0.2 outside attention; attention dropout is 0. The epoch budget is matched; parameter counts and compute costs differ.

2. Part I: prediction quality, cost and learning

Table 3. Test prediction and cost. Lower CE/PPL is better.

Model	Test CE	Test PPL	Best / run epochs	Train wall s	File MiB	128 prefixes s
n-gram	4.6932	109.20	-	0.34 + 0.35*	2.485	0.6905
RNN	4.4222	83.28	7 / 8	111.54	3.337	0.0911
LSTM	4.3262	75.66	8 / 8	199.88	3.673	0.0870
Transformer	4.5679	96.34	6 / 8	172.20	3.784	0.1389

* n-gram: count fitting + validation search, reported separately. Neural wall time covers the training routine, including validation and checkpoint work; it excludes data preparation and final evaluation. File size is serialized model/checkpoint bytes divided by 2^20, not peak RAM.

2.1 Prediction and efficiency tradeoffs

LSTM has the lowest test PPL, followed by RNN, Transformer and n-gram. LSTM reduces PPL by about 9.2% relative to RNN, while its recorded training wall time is about 79% higher. The n-gram fit is inexpensive, but its limited context and sparse counts coincide with the weakest test prediction here. These observations describe these implementations and settings, not universal model-family rankings.

The experiments run on Windows with Python 3.11.15, PyTorch 2.14.0+cpu, four PyTorch threads, and an Intel64 Family 6 Model 141 CPU (16 logical processors reported). The common inference workload is the first 128 test targets with up to 32 preceding tokens, batch 1, full-vocabulary softmax and target-probability extraction. Eight prefixes warm up the model; the median of three repeats is reported. Tensor construction and transfer are included; checkpoint loading is excluded.

Each prefix is recomputed without recurrent-state or KV caching. This is teacher-forced next-token scoring, not autoregressive generation throughput. Baseline and Transformer timings come from different sessions: the 0.0870 vs 0.0911 s LSTM/RNN difference is too small to establish a reliable speed ranking. The Python full-vocabulary n-gram loop is also implementation-specific.

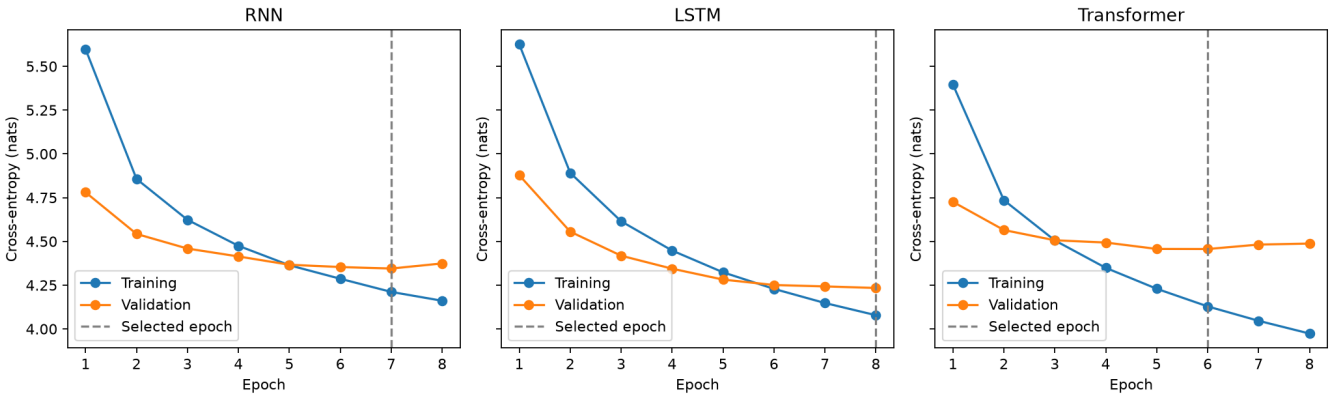


Figure 1. Online training and validation CE for the three neural models; identical epoch cap.

2.2 What the curves support

RNN selects epoch 7 and its validation CE increases at epoch 8. Transformer selects epoch 6, followed by higher validation losses at epochs 7 and 8. These patterns are consistent with overfitting or optimization variation, rather than proof of either cause. LSTM selects epoch 8 while validation CE is still decreasing; the budget has ended, but convergence has not been demonstrated.

All neural models see full paragraphs, unlike the two-token n-gram context. Parameter counts, architectural inductive biases, dropout placement and effective optimization differ. Consequently, a lower LSTM PPL cannot be attributed solely to its gates, and the Transformer result does not show that attention is inherently inferior. Generation provides a different, qualitative test (Section 4).

3. Part II: input embeddings and a freezing control

The Part I LSTM architecture, 100-dimensional inputs, data, optimizer, dropout and training budget remain fixed. Its random/trainable checkpoint is reused; two new LMs are trained from scratch in their remaining layers. The entire input table is frozen in both pretrained settings, including missing and special rows.

3.1 Sources, alignment and separate costs

Corpus-trained embeddings use Skip-gram negative sampling on training paragraphs only: symmetric window 5, five negatives from frequency^0.75, batch 1,024, Adam 0.003, five epochs and seed 42. There is no subsampling or cross-paragraph pairing. Negative draws allow replacement and accidental positives. The 1,117,958 pairs cost 1.92 s to construct; embedding training costs 16.85 s separately from LM training. Only center vectors for ordinary words are transferred; the SGNS objective is not LM cross-entropy.

Public GloVe 6B 100d uses Wikipedia 2014 and Gigaword 5, adding external data. The checksum-verified Gensim-data redistribution changes file format, not the vector source. Exact lowercase matching aligns vectors; missing and special rows share seeded N(0,1) fallback values across the two frozen tables, with PAD zero. No vector normalization is applied. Public pretraining cost is unknown and excluded.

Table 4. Coverage. Type denominator: 3,616 ordinary LM words. Token denominator: all original body tokens.

Source	Matched types	Train tokens	Val. tokens	Test tokens
Skip-gram	3,616 (100.00%)	97.99%	97.01%	96.79%
GloVe	3,531 (97.65%)	97.31%	96.52%	96.21%

Table 5. LSTM variants; all ran eight epochs and selected epoch 8. RF is exploratory.

Inputs	Test CE	Test PPL	Trainable params	LM wall s
Random trainable	4.3262	75.66	946,740	199.88
Skip-gram frozen	4.4285	83.81	584,740	171.31
GloVe frozen	4.4202	83.11	584,740	173.46
Random frozen (RF)	4.4531	85.90	584,740	182.89

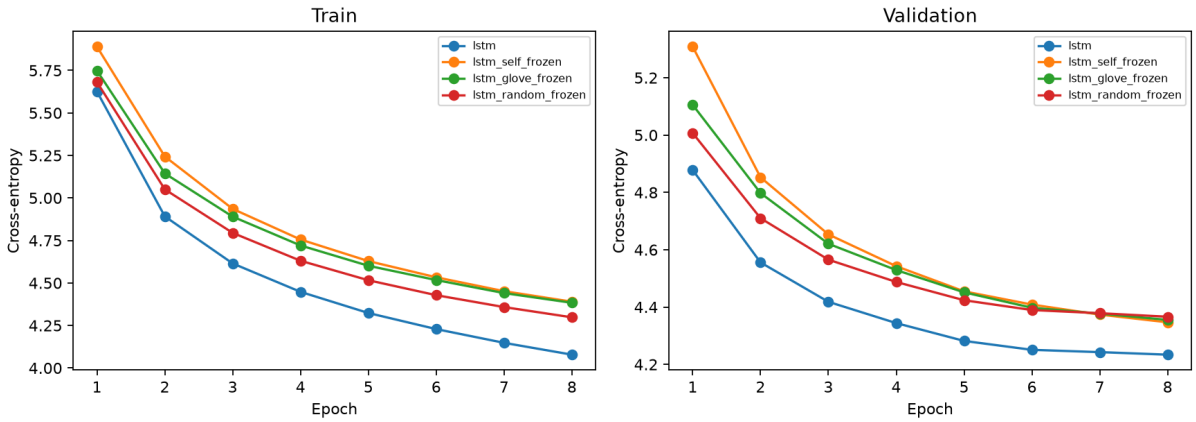


Figure 2. The shared LSTM with four input settings. Frozen tables remain bitwise unchanged.

The original three-way comparison favors random/trainable inputs (75.66 PPL); GloVe and Skip-gram are close (83.11 vs 83.81). Skip-gram has slightly better validation CE than GloVe, but worse test CE. One seed does not establish a stable ordering. All three still improve at epoch 8; losses and checked gradients remain finite, which supports numerical stability only.

An exploratory random-frozen control recreates the original seed-42 initialization and changes only input trainability. Its PPL is 85.90: allowing updates helps this matched pair. Both pretrained-frozen tables perform better than random-frozen, but not better than random/trainable. Vector scale, coverage, pretraining objective and external data remain confounded. Timing across sessions does not isolate a speed benefit.

4. Generated text: matched conditions and concrete failures

All seven LM settings are compared using prompts *elizabeth was*, *mr darcy* and *she could not*, paired with seeds 42/43/44, temperature 0.8 and top-k 50. Each produces exactly 50 new tokens. PAD, BOS and EOS are blocked to hold length constant; UNK is allowed. This sampling distribution differs from the unmasked distribution scored by CE/PPL. Neural models run in evaluation mode and recompute the supplied history.

Table 6. Unedited 24-token prefixes for the same prompt: elizabeth was (seed 42).

Model	**Continuation excerpt**
n-gram	at the same time than any relations , with the , to be the other table , , who had and he had been ...
RNN	all in the course of <unk> on the whole party , and the <unk> of her uncle . but i think of his own ...
LSTM	pleased to the subject . elizabeth to be the <unk> of his own , that the whole party have been seen to her . ...
Transformer	convinced that she had the former in the subject of the evening , and had been a family of the lodges of the <unk> ...

The n-gram combines familiar phrases with malformed transitions and repeated commas. RNN produces recognizable dialogue and later *in love with him*, but also the broken phrase *be to be*. LSTM has the best PPL yet begins *pleased to the subject* and later repeats *and and*. Transformer opens naturally with *convinced that she had*, then drifts into *had been a family of the lodges*. A locally fluent opening is not coherent narration.

Table 7. Unedited 24-token prefixes for she could not (seed 44); identical LSTM architecture.

Input setting	**Continuation excerpt**
Random trainable	help repeating for the other , and the <unk> of others was the country ; and , at home , and the housekeeper , ...
Skip-gram frozen	help it from the whole of the room , and for his father , , to the pleasure of the <unk> of seeing her ...
GloVe frozen	have been some to be <unk> , and the other passed with the room of the <unk> . elizabeth was not to such a ...
Random frozen	have been known to her , and the <unk> of wickham's to be of <unk> . " they had been <unk> to jane , ...

Random/trainable and Skip-gram inputs start with plausible *help repeating* and *help it*, but the latter later generates *her father did not have been great*. GloVe begins *have been some to be*, and random-frozen later says *had been in a servant*. Pretrained vectors do not ensure grammatical continuations. In the separate *mr darcy* samples, both pretrained-frozen models immediately continue with *bennet*, mixing character identities.

The tables report the first 24 tokens of each continuation using a fixed excerpt rule. All complete 50-token sequences and token IDs remain in the saved generation files; repeated sampling with the same seeds reproduces the IDs. Three prompts per setting illustrate repetition, local fluency and consistency failures, but cannot establish a population-level generation-quality ranking. Unknown-token emission and blocked EOS further constrain interpretation. Lower PPL does not guarantee that every sampled passage is better.

5. Part III: frozen representations for sentiment

UCI Sentiment Labelled Sentences supplies 3,000 positive/negative sentences from Amazon, IMDb and Yelp. Deduplicating original tokenizer outputs before UNK mapping removes 21 duplicates, leaving 2,979; no conflicting-label group is found. Seed-42 source-and-label stratification gives 1,787 train, 596 validation and 596 test sentences. The test set has 298 of each class. Original token keys are disjoint across splits.

5.1 Representation and classifier protocol

The three Part II LSTMs are restored with every LM parameter frozen. A sentence becomes BOS+body+EOS under the original novel vocabulary. Mean pooling aggregates 128-dimensional contextual LSTM states at body positions, excluding BOS/EOS/PAD positions, without truncation. UNK positions remain included; BOS may affect later states. These are contextual features before the vocabulary projection, not averages of raw embeddings.

All three representations use StandardScaler plus logistic regression: L-BFGS, maximum 2,000 iterations, seed 42, C in {0.1,1,10}. Scaling and classification are fitted on training data only; C is selected by validation macro-F1, breaking ties with smaller C. All select C=0.1. The full pipeline is refitted on train+validation (2,383 sentences), then evaluated on the test set. No LM is fine-tuned.

Table 8. Same 596 test sentences. F1 intervals are pointwise 95% paired-bootstrap intervals.

Representation	LM PPL	C	Accuracy	Macro-F1	F1 interval
Random/trainable LM	75.66	0.1	0.6326	0.6325	[0.592, 0.671]
Skip-gram/frozen LM	83.81	0.1	0.6359	0.6357	[0.596, 0.673]
GloVe/frozen LM	83.11	0.1	0.6862	0.6861	[0.648, 0.723]
TF-IDF (exploratory)	-	10	0.8255	0.8254	[0.794, 0.854]
Majority reference	-	-	0.5000	0.3333	-

The descriptive LM ranking (random, GloVe, Skip-gram; lower PPL first) differs from the downstream ranking (GloVe, Skip-gram, random; higher F1 first). This shows different observed orderings, not a reliable reversal across populations or seeds. GloVe has the highest observed source-level F1 on all three sources; the next page shows why its aggregate lead does not mean every sentence improves.

5.2 Exploratory lexical baseline and paired uncertainty

TF-IDF uses saved original tokens before novel-UNK mapping, word unigrams/bigrams, min_df=2, a 50,000-feature cap, sublinear TF and L2 normalization, without dense standardization. Vocabulary/IDF are train-fit during the same C-grid search; the entire pipeline is then refit on train+validation. C=10 is selected, at the grid boundary; larger C values remain untested. The final 5,399 features yield 492/596 correct predictions, versus 409 for GloVe features. Task vocabulary, feature dimension and normalization differ, so the comparison does not isolate neural versus lexical architecture.

Uncertainty is estimated with 10,000 paired sentence bootstrap resamples (seed 4213), holding fitted models fixed. Skip-gram minus random accuracy is +0.34 percentage points, with 95% interval [-4.36, 5.03]. GloVe minus random is +5.37 points, interval [1.01, 9.73]. Exact two-sided McNemar tests, Holm-adjusted across all six model pairs, give p=0.0639 for GloVe against either LM alternative. These do not meet family-level 0.05 significance; pointwise intervals are not simultaneous intervals.

TF-IDF minus GloVe accuracy is +13.93 points, interval [9.90, 17.95], with adjusted p=1.50e-10; TF-IDF also exceeds both other LM classifiers after correction. This is stronger evidence conditional on the fitted models. Bootstrap omits training/split variability and assumes independent sentence units.

6. Failures, lessons and reproducibility

6.1 Representative sentiment errors

Table 9. Deterministically selected high-confidence errors; 0=negative, 1=positive.

LM features	Sentence (verbatim)	True / pred.	P(positive)
Random	His losing his marbles so early in the proceedings is totally implausible and unconvincing.	0 / 1	0.903
Random	Not too screamy not to masculine but just right.	1 / 0	0.023
Skip-gram	say bye bye to your tip lady!	0 / 1	0.942
GloVe	I dressed up to be treated so rudely!	0 / 1	0.946
GloVe	Def coming back to bowl next time	1 / 0	0.039

Each LM's saved error set selects the most-confident false positive and false negative, with ID tie-breaking; Table 9 omits the repeated Skip-gram false-negative sentence. These are diagnostic extremes, not a random error sample, and probabilities are uncalibrated. In the first row, *implausible* and *unconvincing* map to UNK; in the GloVe false positive, so does *rudely*. Lost polarity cues are a plausible contributor. Negation and the implied intention to return also challenge mean-pooled features plus a linear classifier; none of these mechanisms is causally isolated.

The three LM classifiers disagree on 285 test sentences. GloVe correctly classifies the negative *causing discomfort* example that the others miss, but incorrectly rejects the positive *The vegetables are so fresh and the sauce feels like authentic Thai.* that both others get right. GloVe's confusion counts are TN=211, FP=87, FN=100, TP=198; TF-IDF gives 252, 46, 58, 240.

6.2 Main conclusions and limits

Prediction: LSTM is the best tested novel LM, with a training-cost tradeoff against RNN. **Embeddings:** jointly trainable inputs outperform the frozen alternatives; the matched random-frozen control supports a benefit from input updates in this run. Pretrained frozen tables partially recover the loss from random freezing. **Transfer:** novel PPL does not reliably determine sentiment utility; GloVe features have the highest observed LM-feature score, while TF-IDF is the stronger practical baseline.

Limits include one novel, one optimization seed, an eight-epoch cap, unequal parameter counts, and only three generation prompts. GloVe adds external data and unequal vector scales. Novel vocabulary produces 21.28/21.41/21.71% UNK on downstream train/validation/test. Although original token keys are disjoint, some different sentences collapse to the same encoded sequence (2 train-validation, 3 train-test and 3 validation-test collisions). All sources occur in every split; unseen-domain transfer is not evaluated, and author/review grouping is unavailable.

Random-frozen training, TF-IDF and uncertainty analysis were designed after examining the original test results. They are exploratory: validation-only parameter selection does not undo test-informed research design. Confirming the findings would require fresh held-out data and repeated optimization seeds.

6.3 Evidence and reproducibility

The English Notebook contains model definitions, training and evaluation; README records data access, environment, configurations and seeds. Saved JSON/CSV results, full generations, classifier probabilities and checkpoints support the tables. All 51 code cells passed a sequential check in an isolated copy, restoring LM/Skip-gram checkpoints in place of retraining. This verifies the checkpoint-based workflow, rather than a fresh training run. Run All currently rewrites results and mixes training with one cached control, so the saved artifacts are retained alongside the code.

Data and Code Sources

Data, pretrained vectors and software repositories used:

Project Gutenberg: Pride and Prejudice (ebook 1342)
https://www.gutenberg.org/ebooks/1342

UCI: Sentiment Labelled Sentences (Kotzias, 2015; CC BY 4.0)
https://doi.org/10.24432/C57604

Stanford GloVe
https://github.com/stanfordnlp/GloVe

gensim-data: glove-wiki-gigaword-100 vectors
https://github.com/piskvorky/gensim-data

PyTorch
https://github.com/pytorch/pytorch

scikit-learn
https://github.com/scikit-learn/scikit-learn

SciPy
https://github.com/scipy/scipy

NumPy
https://github.com/numpy/numpy

Matplotlib
https://github.com/matplotlib/matplotlib

ReportLab
https://github.com/MrBitBucket/reportlab-mirror

AI assistance supported report formatting, information lookup and code auditing. The tables present saved experimental results, and the figures are the original color plots exported by the Notebook.